

Can one hear an aperiodic tile?

Spectral geometry, planar tiling and forced aperiodicity

A systematic research study prepared with Codex

7 October 2026

Main finding. An isolated tile's ordinary Laplacian eigenvalues are not currently a general certificate of forced aperiodicity. There are exact obstructions from missing placement rules, and a provable instability even for unmarked geometric monotiles. A much richer billiard invariant can recover many polygonal shapes; spectra of the entire tiling can detect long-range order under explicit assumptions.

1. Questions and study goals

I chose five goals: define the spectral objects precisely; locate the strongest tiling theorems; distinguish impossibility results from unresolved inverse problems; investigate a family crossing between periodic tileability and forced aperiodicity; and identify what could become a useful certificate for tiling search. The literature study, analytic deductions and numerical experiment are completed here. A follow-up adds planar and spatial chair controls, including the recent Chair44 construction. The remaining questions are identified as research problems, not presented as solved.

2. Fix the object before asking the question

A geometric tile is a bounded planar domain Ω with a specified group G of allowed isometries. A tiling covers the plane with copies $g_i\Omega$, with disjoint interiors. For measurable translational tilings, coverage and disjointness are understood almost everywhere. For marked tiles, matching rules are additional data.

A tiling $\mathcal{T}$ has a translational period if $\mathcal{T} + v = \mathcal{T}$ for a nonzero vector v . A fully periodic tiling has two independent periods. In this study, a **strongly aperiodic tiling** has no nonzero translational period. A **forced-aperiodic tile** admits at least one tiling and every admitted tiling is strongly aperiodic. Merely constructing one nonperiodic arrangement does not establish this universal claim. Both the allowed group and the matching rules must be fixed.

Data	What is measured?	Relationship to forced aperiodicity
Laplacian spectrum	Eigenvalues of an operator on the isolated tile, with chosen boundary conditions.	Geometric information and necessary inequalities; no general test established here.
Billiard length spectrum	Lengths of closed reflecting trajectories; marked versions retain trajectory labels.	Reflection geometry, with inverse results depending strongly on retained labels.
Bounce spectrum	The complete language of labeled boundary encounters.	Recovers non-right-angled polygons up to similarity; hence can determine geometric properties indirectly.
Fourier spectral set	An orthogonal basis of plane waves on the tile.	Equivalent to translational tileability for convex bodies; lattice cases certify a periodic option.
Diffraction spectrum	Fourier transform of the autocorrelation of a chosen decorated point pattern.	Detects long-range correlations of that pattern, with information loss.
Dynamical spectrum	Translation action on the space of legal tilings.	Can test periods if suitable eigenfunctions separate configurations.

The first four start with a tile; the last two normally require a tiling or a tiling space. Confusing these levels would turn a statement about a known arrangement into an unsupported claim about every possible arrangement.

3. The strongest direct bridge: Fourier spectra

A measurable domain is Fourier spectral if some countable frequency set Λ makes the normalized exponentials an orthonormal basis:

$$e_\lambda(x) = |\Omega|^{-1/2} \exp(2\pi i \lambda \cdot x), \quad \lambda \in \Lambda.$$

These are not generally Dirichlet or Neumann eigenfunctions on Ω . Their orthogonality means

$$\widehat{\mathbf{1}_\Omega}(\lambda - \mu) = 0 \quad (\lambda \neq \mu),$$

together with a completeness condition. Fuglede's conjecture linked this property to tiling by translations. For convex planar domains it is a theorem: spectrality is equivalent to translational tiling, and the tiles are parallelograms and centrally symmetric hexagons. Iosevich–Katz–Tao proved the planar result; Lev–Matolcsi later proved the convex-body equivalence in every dimension. [1] [2]

The lattice case has a transparent mechanism

Let L be a full-rank lattice and L^* its reciprocal lattice. Periodize the indicator:

$$P_L(x) = \sum_{\ell \in L} \mathbf{1}_\Omega(x - \ell), \quad L^* = \{k : k \cdot \ell \in \mathbb{Z} \text{ for every } \ell \in L\}.$$

Its Fourier coefficients are $\widehat{\mathbf{1}_\Omega}(k)/\text{covol}(L)$. Therefore the exact conditions

$$|\Omega| = \text{covol}(L), \quad \widehat{\mathbf{1}_\Omega}(k) = 0 \quad (k \in L^* \setminus \{0\})$$

are equivalent to $P_L = 1$ almost everywhere. They certify a lattice tiling and thus exclude forced aperiodicity for any allowed-placement model containing those translations. Testing finitely many Fourier zeros numerically does not certify this infinite set of conditions. A Fourier basis indexed by a reciprocal lattice supplies a periodic option; an arbitrary nonlattice Fourier spectrum does not automatically do so.

Every triangle can tile using appropriate congruent copies: a triangle and its half-turn make a parallelogram, which repeats. Yet a convex triangle is not Fourier spectral, by the planar classification. Thus “not Fourier spectral” does not imply “cannot tile” when rotations are permitted. [1]

Current literature status. The unrestricted Fuglede equivalence fails in higher dimensions. A July 2026 preprint by Tao Zhang claims counterexamples to both directions already in the plane. I checked its statement and constructions at the source but have not independently validated its proof; this study does not rely on that claim. The established convex theorem and the lattice argument remain the relevant safe conclusions. [3] [4]

4. What the Laplacian does know about tiles

For the Dirichlet drum, solve $-\Delta u = \lambda u$ in Ω with $u|_{\partial\Omega} = 0$; list eigenvalues $0 < \lambda_1 \leq \lambda_2 \leq \dots$, including multiplicities. For Neumann conditions, $\partial_\nu u = 0$, the connected-domain spectrum begins $0 = \mu_0 < \mu_1 \leq \dots$. Scaling by s multiplies either spectrum by s^{-2} ; reflection and rotation leave it unchanged.

For a straight-sided polygon with interior angles θ_i , area A and perimeter P , the Dirichlet heat trace has the expansion

$$H_D(t) = \sum_{j \geq 1} e^{-t\lambda_j} = \frac{A}{4\pi t} - \frac{P}{8\sqrt{\pi t}} + C_\Omega + O(e^{-c/t}), \quad C_\Omega = \sum_i \frac{\pi^2 - \theta_i^2}{24\pi\theta_i}.$$

Thus the spectrum determines these three geometric quantities. All further coefficients at algebraic orders vanish for such polygons; the full heat trace contains additional information beyond this expansion. In the class of triangles, area, perimeter and the reciprocal-angle sum determine the shape, giving a positive inverse spectral theorem. [5]

For general planar domains, shape recovery fails: Gordon–Webb–Wolpert constructed noncongruent domains with identical full Dirichlet spectra. **That fact alone does not prove that tiling or aperiodicity differs within an isospectral pair.** Such a conclusion would require a pair with separately established, different tiling properties. I did not locate a verified geometric pair with one member forcing aperiodicity and the other admitting periodic tilings. [6]

A genuine tiling-to-eigenvalues theorem

Define $N_D(E) = \#\{j : \lambda_j < E\}$ and $N_N(E) = \#\{j : \mu_j < E\}$, including the Neumann zero mode. For planar tiling domains with the regularity needed for the operators, Pólya’s inequalities give

$$N_D(E) \leq \frac{AE}{4\pi}, \quad N_N(E) \geq \frac{AE}{4\pi} \quad (E > 0).$$

The mechanism is to fill a large region with copies of the tile and compare operators by boundary bracketing, then use Weyl asymptotics. Filonov supplies the general Neumann tiling result. The argument does not require that the chosen arrangement be periodic, so it cannot separate periodic tiles from forced-aperiodic ones. [7]

The converse fails even as a tileability test: the disk obeys both inequalities but cannot tile the plane by congruent copies. A certified violation could reject tileability under the theorem's hypotheses; satisfying the inequalities does not establish it. [8]

5. Geodesics, reflections and symbolic rigidity

A flat domain's ordinary interior geodesics are straight lines. There is no useful collection of nonconstant closed interior geodesics. One must specify reflecting billiards, a doubled surface, or a surface formed by a prescribed edge gluing. Those last two constructions include data that the isolated tile does not supply on its own.

In a rectangle of side lengths a, b , unfolding reflections gives closed-orbit lengths of the form $2\sqrt{m^2a^2 + n^2b^2}$, with admissible integer pairs and repetitions treated explicitly. Its Dirichlet spectrum is $\pi^2(m^2/a^2 + n^2/b^2)$ for positive integers. This common lattice geometry explains the relation in a special case; it is not a universal tiling criterion.

Rational polygon billiards unfold to compact translation surfaces, usually with cone singularities. The unfolded copies can overlap when projected to the plane. Compact unfolding therefore does not prove a plane tiling. Rational polygons include examples with quite different tiling behavior; rationality alone does not detect forced aperiodicity. [9]

The wave trace $\sum_j \cos(t\sqrt{\lambda_j})$, interpreted as a distribution, links eigenvalues to lengths of billiard trajectories. Corners introduce diffraction, and cancellations and degeneracies matter. One must not identify wave-trace singularities with a complete, labeled catalogue of all trajectories. Unmarked lengths discard which sides were encountered.

A positive answer with richer data. Duchin–Erlandsson–Leininger–Sadanand show that two simply connected Euclidean polygons with the same complete bounce spectrum are similar, except for the right-angled affine exception. The bounce spectrum records all bi-infinite sequences of side labels along nonsingular trajectories. For a non-right-angled polygon, it determines the shape up to similarity. With the placement rules fixed, the tiling and forced-aperiodicity properties are consequently determined as geometric properties of that recovered shape. This is an inference from their rigidity theorem, not a practical finite test or a theorem about unmarked lengths. [10]

This applies to the polygonal Hat–Turtle family because its angles are not all multiples of $\pi/2$. It does not extend directly to curved-edge Spectres. Recovering a shape also does not, by itself, provide a terminating algorithm to decide every tiling problem about that shape.

6. Three limits on detecting forced aperiodicity

Placement rules are invisible to an ordinary drum

The equilateral polygon $\text{Tile}(1, 1)$ has periodic tilings if reflected copies are permitted. With reflections forbidden, its admitted tilings are all aperiodic. This is the weakly chiral monotile result. The physical domain and both its ordinary Dirichlet and Neumann spectra are identical in these two problems.

Any detector using only those eigenvalues must give the same answer, even though the correct forced-aperiodicity answer changes. A detector must therefore accept the placement group as additional input. [11]

Matching rules are also invisible unless coupled to the operator

Adding arrows or colors to an unmodified tile boundary leaves a homogeneous drum unchanged. For example, the two Penrose rhombi with their matching rules force aperiodicity; without those rules each rhombus is a parallelogram admitting lattice tilings. The catalog of isolated shapes and its Laplacian spectra stay the same. This is an obstruction for decorated tile systems, not an isospectral counterexample between two different unmarked monotile shapes. [12]

Even with fixed rules, finite noisy spectra are unstable

Let $\Omega_r = \text{Tile}(1, r)$, with all planar isometries allowed. The established family result gives periodic tilings at $r = 1$, whereas every positive $r \neq 1$ forces aperiodicity. [13] The boundary coordinates vary continuously. In a neighborhood of $r = 1$, a fixed triangulation gives a piecewise affine, bi-Lipschitz map from Ω_1 to Ω_r , tending to the identity.

Deduction in this study Pull back the energy and mass forms through this map. Their Jacobians converge uniformly on the finitely many triangles. The min–max characterization then gives, for every fixed index j ,

$$\lambda_j(\Omega_r) \longrightarrow \lambda_j(\Omega_1), \quad \mu_j(\Omega_r) \longrightarrow \mu_j(\Omega_1) \quad (r \rightarrow 1).$$

Area normalization preserves this convergence. Consequently, for any finite number K of modes and any positive measurement tolerance ε , a forced-aperiodic member can satisfy

$$\max_{1 \leq j \leq K} |A_r \lambda_j(\Omega_r) - A_1 \lambda_j(\Omega_1)| < \varepsilon.$$

The same statement holds for a finite combination of Dirichlet and Neumann modes. Their measurement intervals can overlap those of a periodically tileable member. Thus there is no universally robust finite-noise classifier over this family. This does **not** rule out a classifier using exact finite data in a restricted parametric family, or a decision theorem from the full exact infinite spectrum. Near equality is not isospectrality.

7. An exact counterexample for heat invariants

Analytic deduction, not a full isospectral pair There is a stronger failure for the usual heat invariants. For this family the angles are fixed, and direct geometry gives

$$A(r) = \sqrt{3}(2 + \sqrt{3}r + r^2), \quad P(r) = 8 + 6r, \quad C(r) = \frac{11}{36}.$$

Choose a second ratio and scale it to the perimeter of the equilateral member:

$$r_* = \frac{50\sqrt{3} - 48}{22\sqrt{3} - 27} \approx 3.476103648, \quad s_* = \frac{14}{8 + 6r_*} \approx 0.485157273.$$

The equation $P(r)^2/A(r) = P(1)^2/A(1)$ has exactly the roots 1 and r_* . Indeed, after clearing the denominator it factors as

$$(108 - 88\sqrt{3})(r - 1)(r - r_*) = 0.$$

It follows that Ω_1 and $s_*\Omega_{r_*}$ have exactly equal area, perimeter and corner heat constant. One admits periodic tilings; the other forces aperiodicity. Their **entire algebraic small-time Dirichlet heat expansions agree**, since a polygon has only the three nonzero coefficients displayed above. The exponentially small remainder and the full spectra need not agree.

This rules out forced-aperiodicity detection from those heat coefficients, even with exact measurements and a fixed group allowing all isometries. It illustrates why hearing corner complexity, or recovering a few global geometric invariants, is insufficient. The calculation is independently checked in the accompanying verification receipt.

8. A numerical investigation of the transition

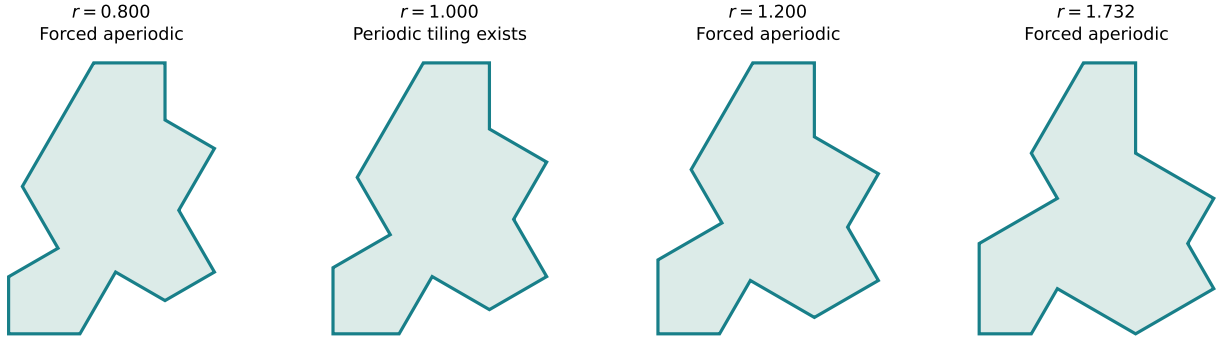


Figure 1: Shapes in the same geometric family. The labels are established by tiling theorems, not inferred from the eigenvalues. All isometries, including reflections, are allowed.

The computation estimates the first twelve Dirichlet eigenvalues at ten parameter values. It uses conforming linear finite elements, a consistent mass matrix, a constrained quality triangulation, a common deformation of that mesh, and three nested refinements. The reported feature is $A_r\lambda_j$, which removes overall scale. The same isolated tile, without an added potential or matching-rule boundary condition, is used throughout.

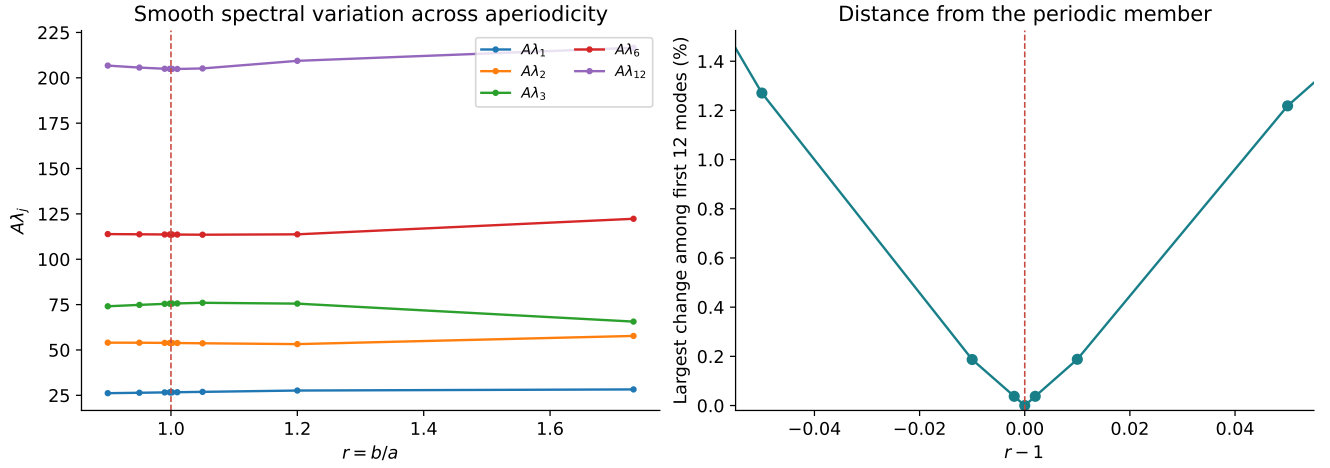


Figure 2: The red line marks the equilateral parameter, where periodic tilings are possible. Nearby parameter values force aperiodicity. The smooth numerical variation illustrates the min-max continuity deduction; it is not its proof.

Ratio	Known tiling property	Largest relative change in first twelve modes
0.9	forces aperiodicity	3.092%
0.95	forces aperiodicity	1.271%
0.99	forces aperiodicity	0.1871%
0.998	forces aperiodicity	0.0375%
1	admits periodic tilings	0%
1.002	forces aperiodicity	0.03753%
1.01	forces aperiodicity	0.188%
1.05	forces aperiodicity	1.219%
1.2	forces aperiodicity	4.045%
1.73205	forces aperiodicity	13.11%

Each finest family mesh has 17065 nodes and 33664 triangles. The largest last-refinement change among the twelve modes is 1.231% over all cases, and 0.409% for $0.95 \leq r \leq 1.05$. The finest square control differs from its exact first twelve modes by at most 0.106%. The largest reported relative eigensolver residual is 2.86×10^{-11} .

Refinement differences measure sensitivity to discretization; they are not rigorous error bounds. A tiny algebraic eigensolver residual is also not a bound on the continuum eigenvalue error. Common meshes help isolate the variation with the parameter, but do not eliminate discretization error. In particular, the small spectral differences near the equilateral member are numerical illustrations, while the analytic continuity argument establishes arbitrary closeness independently.

The experiment has no trained classifier and no inference of labels from a finite patch. It does not claim to discover a new monotile or to independently reprove the Hat-family aperiodicity theorem.

8a. The L-triomino and Chair44: a controlled comparison

The first version omitted this important example. The planar L-triomino, its extrusion, and the three-dimensional seven-cube chair provide three distinct controls. Chair44 adds a fourth: the chair with a

hierarchy-enforcing marking, realized geometrically by bumps and dents. This separates the shape, the matching rule, and its physical realization instead of treating them as one spectral object.

The bare chairs have exact periodic and Fourier certificates

For dimension $d = 2$ or $d = 3$, take the unit-cube cells

$$B_d = \{0, 1\}^d \setminus \{(1, \dots, 1)\}, \quad q_d = 2^d - 1, \quad L_d = \left\{ z \in \mathbb{Z}^d : \sum_{j=1}^d 2^{j-1} z_j \equiv 0 \pmod{q_d} \right\}.$$

The weighted binary sums of the cells in B_d are precisely $0, \dots, q_d - 1$. Thus they give one representative of every coset of L_d , and translating the chair by L_d covers each integer cube exactly once. This proves periodic tileability by translations alone. The planar L-triomino has area 3; the spatial chair has volume 7. Both are nonconvex Fourier spectral sets, with their reciprocal lattices as spectra. This follows from the exact lattice argument in Section 3; convexity is unnecessary in this special case.

The bare chair can also support hierarchical nonperiodic tilings. Its isolated drum spectrum is the same whichever arrangement is chosen. Chair44's decisive feature is that its matching rules require every legal tiling to follow a hierarchy. Goodman-Strauss explains the marked construction and its bump-and-dent realization. The spatial chair is a cube with one corner cube removed; it is not the extrusion of the planar L-triomino. [\[19\]](#)

Planar and extruded drum controls

For the unit L-triomino, direct geometry gives

$$A = 3, \quad P = 8, \quad C = \frac{5}{16} - \frac{5}{144} = \frac{5}{18}.$$

The corner calculation uses five angles of $\pi/2$ and one reflex angle of $3\pi/2$. Adding abstract arrows to this domain leaves these quantities and both homogeneous Laplacian spectra unchanged. A marking that is only a label needs to be supplied separately to a spectral test.

If $E_h = \Omega_L \times (0, h)$ is the L-triomino prism, separation of variables gives the exact product identities

$$\text{Spec}_D(E_h) = \left\{ \lambda_m(\Omega_L) + \frac{\pi^2 n^2}{h^2} : m, n \geq 1 \right\}, \quad \text{Spec}_N(E_h) = \left\{ \mu_m(\Omega_L) + \frac{\pi^2 n^2}{h^2} : m, n \geq 0 \right\},$$

with multiplicities retained. This is an analytically controlled three-dimensional benchmark, not a formula for the seven-cube chair. Repeating a planar periodic tiling and stacking the prisms gives a fully periodic spatial tiling. A planar nonperiodic arrangement stacked identically still has a vertical period, so extrusion alone does not force spatial aperiodicity.

The L-triomino control estimates $\lambda_1 = 9.64657652$ on a finest mesh of 18817 nodes and 37120 triangles. The largest last-refinement change among twelve modes is 0.344%. These are conforming FEM estimates without certified continuum error bounds. The prism estimates use the exact product identity with numerical planar inputs. The [chair control receipt](#) also records exact lattice and parameter checks; the [control script](#) reproduces them.

Physical markings change the spectrum, but can become arbitrarily shallow

Tsiokos's September 2026 preprint specifies a geometric Chair44 with 192 square-pyramid features, distributed among twelve balanced height groups. Its Appendix E.6 states sufficient open conditions for a twelve-parameter family with the same tiling conclusion. This study checks the parameter algebra; it does not independently verify the full aperiodicity proof. [20]

Deduction from the stated parameter theorem Keep the feature bases and centers fixed, and set the component coefficients to $c_j = \delta j$ for $j = 1, \dots, 12$ and $0 < \delta \leq 1$. The height of group j is then $\delta j/10000$. The only potentially nonmonotone angle exclusion becomes

$$i^2 - 2j^2 \neq \frac{\delta^2 j^4}{10000} \quad (1 \leq i, j \leq 12).$$

A negative left side cannot equal the positive right side. For every positive left side, exact rational enumeration verifies $i^2 - 2j^2 - j^4/10000 > 0$; zero left sides are impossible for positive integers. The remaining size inequalities improve as δ decreases, and distinct nonzero magnitudes remain distinct. Consequently the preprint's stated assumptions hold throughout this shallow-feature family. Its theorem supplies forced aperiodicity for every positive δ ; at $\delta = 0$, the bare periodic chair returns.

In the fixed feature tubes, write the boundary deformation as $F_\delta(u, z) = (u, z + \delta\chi(z)f(u))$, with a piecewise linear cutoff χ . The supports are disjoint, and their dimensions stay fixed. Hence the maps are uniformly bi-Lipschitz near zero and their derivatives tend to the identity. Pullback of the energy and mass forms and the min-max principle give

$$\lambda_k(Q_\delta) \longrightarrow \lambda_k(C_3), \quad \mu_k(Q_\delta) \longrightarrow \mu_k(C_3) \quad (\delta \downarrow 0)$$

for each fixed mode, where C_3 is the bare seven-cube chair. This extends the finite-noise obstruction to this spatial family, conditional on the cited Chair44 parameter theorem. It is not a claim that the physical marked solid and the bare chair have identical spectra.

There is also an explicit geometric check. Opposite feature volumes cancel, so $|Q_\delta| = 7$. A square pyramid with base half-width $1/100$ contributes the following surface excess after replacing its flat base:

$$S(Q_\delta) = 24 + \frac{4}{625} \sum_{j=1}^{12} \left(\sqrt{1 + \frac{\delta^2 j^2}{10000}} - 1 \right), \quad 0 < S(Q_\delta) - 24 \leq \frac{13}{62500} \delta^2.$$

Thus volume remains exactly equal to that of the bare chair, while surface area distinguishes every positive feature depth and approaches it quadratically. Two-term heat asymptotics recover volume and surface area from the exact Dirichlet or Neumann spectrum for bounded Lipschitz domains, including these polyhedra. [21] Consequently there is a **positive exact spectral test within this certified family**: if the tile is known independently to be a scaled copy of C_3 or one of the Q_δ , then

$$\frac{S}{V^{2/3}} > \frac{24}{7^{2/3}}$$

distinguishes every positive-depth, forced-aperiodic member from the bare periodic member, using the cited parameter theorem. The equality case is the bare chair. The family-membership assumption is essential: this is not an aperiodicity criterion for arbitrary solids of large surface area. Robust recognition from finitely many noisy modes remains a different question, since the surface excess has no positive lower bound as the features flatten.

What the small example set permits

The most informative benchmark is a controlled sequence: bare L-triomino, its prism with the exact product spectrum, bare seven-cube chair, abstractly marked chair using the same homogeneous operator, and the published geometric Chair44 over decreasing feature depths. Other markings should receive a forced-aperiodicity label only when a theorem covers them. In particular, the project's centered tetrahedral relief is a different geometric realization whose aperiodicity is unproved; it must not inherit the published label automatically.

There are few independent known geometric constructions. Many parameter samples from one construction do not become many independent examples. A classifier evaluated on random samples of the same family may simply estimate feature depth or a geometric parameter. Hold out complete construction families, require proofs for labels, and emphasize exact counterexamples and perturbation arguments. Here the scarcity favors structural tests over claims of statistical universality.

9. Where aperiodicity really appears spectrally

Diffraction combines a tile with its arrangement

For translates in a finite patch, the Fourier amplitude of a density formed by copies of the tile factors as

$$\hat{\rho}(k) = \widehat{\mathbf{1}}_{\Omega}(k) \sum_{x \in \Lambda} e^{-2\pi i k \cdot x}.$$

The first factor is the tile's form factor; the second depends on the arrangement of copies. Rotated tiles require separate orientation contributions. Exact tiling by uniform density has $\rho = 1$ almost everywhere, so it produces only the trivial zero-frequency contribution: **an observation that erases tile boundaries can erase all aperiodicity information**. Useful diffraction needs control points, vertex sets, boundary contrast or another specified decoration.

For a suitable translation-bounded point measure ω , its autocorrelation is the volume-averaged convolution γ , and its diffraction is $\hat{\gamma}$. Sharp Bragg peaks do not imply periodicity. Hat-family tiling dynamics have pure point spectrum, and the diffraction of Hat and Spectre tilings has now been derived through model-set representatives and reprojections. These are spectral results about their long-range tilings, not the isolated drum. [14] [15]

A periodic decorated pattern has diffraction supported on a reciprocal lattice. Incommensurate nonzero peaks can exclude candidate period lattices; a finite diffraction image usually cannot certify exact incommensurability. Extinctions may hide relevant frequencies. Distinct homometric structures share autocorrelation and diffraction, so diffraction generally does not reconstruct all boundary or matching information. [16]

Dynamical eigenfunctions test translation periods

Let X be a specified legal tiling space. A continuous translation eigenfunction obeys

$$f_k(T - v) = e^{2\pi i k \cdot v} f_k(T).$$

If v is a period of T and $f_k(T) \neq 0$, then $k \cdot v \in \mathbb{Z}$. Thus available eigenfrequencies constrain its periods through

$$\text{Per}(T) \subseteq \{v : k \cdot v \in \mathbb{Z} \text{ for all such } k\}.$$

Conditional certificate If this annihilator is trivial, those eigenfunctions certify that the configuration has no nonzero period. If continuous eigenfunctions separate configurations and the condition applies throughout the full legal space, it can certify forced aperiodicity of that space. Measure-theoretic statements about almost every tiling alone do not exclude exceptional periodic tilings. Counting the rank of the frequency module alone also need not exclude a remaining one-dimensional period.

Pure point diffraction and pure point dynamical spectrum are equivalent in the appropriate finite-local-complexity setting with uniform cluster frequencies, retaining the relevant colors or observables. A single observation can lose information. [17] Before this helps the monotile question, one must establish that the studied space includes every legal tiling of that tile.

A Laplacian on the full tiled plane needs an interface model

If all seams are transparent and the operator is simply the Euclidean Laplacian on the whole plane, its spectrum is $[0, \infty)$, regardless of the tiling. If each tile is isolated by Dirichlet walls, the direct-sum spectrum repeats the same tile eigenvalues and forgets its arrangement. Geometry can influence an operator with seam interactions, graph adjacency, varying material coefficients or a tiling-dependent potential, but those are additional models. Band gaps, localized states and level statistics are then model-dependent observations, not automatic certificates of forced aperiodicity.

10. What I would investigate next

The evidence favors **boundary and compatibility information** over a scalar eigenvalue fingerprint. The following priorities each have a concrete success criterion.

1. **Resolve the exact-spectrum question with fixed placement rules.** Seek either an isospectral pair of unmarked planar monotiles with different forced-aperiodicity status, or an inverse theorem for a meaningful class containing known monotiles. A pair must have a certified infinite construction for each tiler and a proof excluding every periodic tiling for the aperiodic member. Noncongruence alone is insufficient. This study establishes no such pair.
2. **Study boundary response and attachment.** Retain a parameterized Dirichlet-to-Neumann operator or boundary eigenfunction data, rather than just eigenvalues. Gluing two boundaries needs continuity of the field and cancellation of outward flux, information scalar spectra omit. Inverse boundary spectral results exist under precise admissibility assumptions; they are a reason to study richer data, not a blanket reconstruction theorem for every tile. [18] Success would be an exact reduction of some boundary incompatibility to an independently checkable spectral certificate.
3. **Exploit complete bounce data in polygonal families.** Use symbolic rigidity to recover geometry in the non-right-angled class, then apply geometric tiling theorems. The target is a restricted reconstruction procedure with a declared completeness guarantee. A finite collection of sampled trajectories must not be treated as the complete bounce language.

4. **Build a full-space dynamical certificate.** Identify a forced hierarchy for every legal tiling, derive its translation eigenfunctions, and show their period annihilator is trivial. A substitution generating some aperiodic examples is insufficient; recognition and exhaustive coverage of legal configurations are the decisive steps.

In a known parametric family such as $\text{Tile}(1, r)$, exact geometry already supplies the aperiodicity answer. Spectral measurements could estimate the parameter and thereby suggest a label, but the guarantee would come from the family theorem. It would not transfer automatically to arbitrary custom tiles.

11. Implications for GCTS and proof scope

This study changes no tiling engine. Its finite-element solver is a continuum spectral experiment, not an implementation of the point-value tiling baseline. I read the project's master algorithm contract; no conformance to its frontier scheduler or candidate graph is claimed for the FEM computation.

For future integration, spectral features may rank candidates within the permitted generation ties or help propose clusters. They must not silently remove candidates. A proved Fourier lattice identity can certify a periodic construction; a proved necessary spectral inequality may justify an exclusion under its hypotheses. A learned spectral correlation remains a hypothesis and cannot turn budget exhaustion into non-tiling or forced aperiodicity.

The base tiling model must still specify the exact point domain, values, markings and allowed transforms; maintain complete point-candidate incidence; check global dead ends and forced moves before earliest-generation branching; preserve exact rollback; and independently verify all occupancy and matching claims. The continuum-to-point reduction itself needs justification. A spectral score does not replace those obligations.

12. Answers to the original questions

- **Can spectra connect to whether a tile tiles?** Yes: Fourier spectrality characterizes convex translational tiles; lattice Fourier conditions certify periodic constructions; Laplacian inequalities are necessary for tiling domains.
- **Can ordinary drum data determine forced aperiodicity?** The answer is no if placement rules or decorations are omitted. Exact heat coefficients fail even with fixed rules. Finite noisy modes have a provable instability. For the full exact infinite Laplacian spectrum of an unmarked geometric tile with rules fixed, this study leaves the general question unresolved.
- **Can geodesic information help?** Complete symbolic billiard data can determine non-right-angled polygons up to similarity, so it determines their geometric tiling properties indirectly. Ordinary unmarked lengths are a different, weaker input.
- **Which spectrum is most relevant to aperiodic order?** The translation dynamics of the complete legal tiling space, supported by specified diffraction observables, has the strongest direct relationship. It still needs a theorem connecting that space to every allowed tiling of the tile.

Primary sources and audit trail

Sources checked on 7 October 2026. Established theorems, a recent preprint claim, deductions in this study and numerical observations are distinguished in the text. The search focused on Fourier tiling, inverse planar spectra, billiard rigidity, Hat/Spectre dynamics and a Chair44 follow-up; this is a systematic thematic study, not an exhaustive bibliometric review.

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2. Lev and Matolcsi. [The Fuglede conjecture for convex domains is true in all dimensions](#). Convex-body theorem.
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Additional verified context: [Bhattacharya's planar lattice translational periodicity theorem](#) says a finite subset of the integer plane that tiles by translations has a fully periodic tiling. It concerns that lattice model, not arbitrary rotations of a Euclidean monotile.